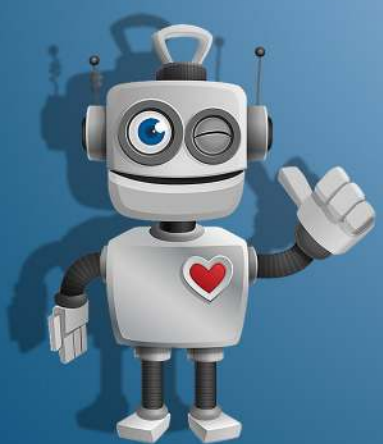
تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

امتحان سابق لمادة

مكونات الألة و آلياتها





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EXAMPLE PROBLEM 1.7

Figure 1.26 presents a lift table used to adjust the working height of different objects. Draw a kinematic diagram and compute the degrees of freedom.

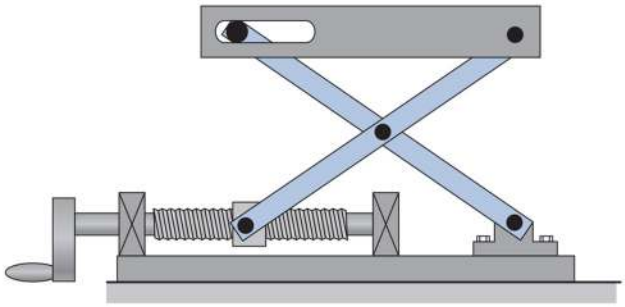


FIGURE 1.26 Lift table for Example Problem 1.7.

SOLUTION:

1. ***Identify the Frame***

   The bottom base plate rests on a fixed surface. Thus, the base plate will be designated as the frame. The bearing at the bottom right of Figure 1.26 is bolted to the base plate. Likewise, the two bearings that support the screw on the left are bolted to the base plate.

   From the discussion in the previous section, the out-of-plane rotation of the screw will not be considered. Only the relative translation of the nut will be included in the kinematic model. Therefore, the screw will also be considered as part of the frame. The motion of all other links will be determined relative to this bottom base plate, which will be numbered as link 1.

2. ***Identify All Other Links***

   Careful observation reveals five other moving parts:

   Link 2: Nut
   Link 3: Support arm that ties the nut to the table
   Link 4: Support arm that ties the fixed bearing to the slot in the table
   Link 5: Table
   Link 6: Extra link used to model the pin in slot joint with separate pin and slider joints

3. ***Identify the Joints***

   A sliding joint is used to model the motion between the screw and the nut. A pin joint, designated as point *A*, connects the nut to the support arm identified as link 3. A pin joint, designated as point *B*, connects the two support arms—link 3 and link 4. Another pin joint, designated as point *C*, connects link 3 to link 6. A sliding joint joins link 6 to the table, link 5. A pin, designated as point *D*, connects the table to the support arm, link 3. Lastly, a pin joint, designated as point *E*, is used to connect the base to the support arm, link 4.

4. ***Draw the Kinematic Diagram***

   The kinematic diagram is given in Figure 1.27.

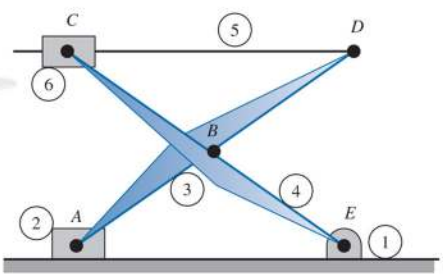


FIGURE 1.27 Kinematic diagram for Example Problem 1.7.

5. ***Calculate Mobility***

   To calculate the mobility, it was determined that there are six links in this mechanism. There are also five pin joints and two slider joints. Therefore

$$n = 6 \quad j_p = (5 \text{ pins } + \text{ 2 sliders}) = 7 \quad j_h = 0$$

   and

$$M = 3(n - 1) - 2j_p - j_h = 3(6 - 1) - 2(7) - 0 = 15 - 14 = 1$$

   With one degree of freedom, the lift table has constrained motion. Moving one link, the handle that rotates the screw, will precisely position all other links in the device, raising or lowering the table.

EXAMPLE PROBLEM 1.8

Figure 1.29 shows a mechanical press used to exert large forces to insert a small part into a larger one. Draw a kinematic diagram, using the end of the handle as a point of interest. Also compute the degrees of freedom.

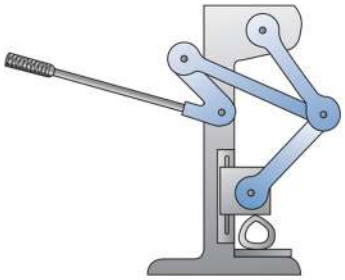


FIGURE 1.29 Mechanical press for Example Problem 1.8.

SOLUTION:

1. ***Identify the Frame***

   The bottom base for the mechanical press sits on a workbench and remains stationary during operation. Therefore, this bottom base is designated as the frame. The motion of all other links is determined relative to the bottom base. The frame is numbered as link 1.

2. ***Identify All Other Links***

   Careful observation reveals five other moving parts:

   Link 2: Handle
   Link 3: Arm that connects the handle to the other arms
   Link 4: Arm that connects the base to the other arms
   Link 5: Press head
   Link 6: Arm that connects the head to the other arms

3. ***Identify the Joints***

   Pin joints are used to connect the several different parts. One connects the handle to the base and is labeled as joint *A*. Another connects link 3 to the handle and is labeled as joint *B*. Another connects link 4 to the base and is labeled as *C*. Another connects link 6 to the press head and is labeled as *D*.

   It appears that a pin is used to connect the three arms—links 3, 4, and 6—together. Because three separate links are joined at a common point, this must be modeled as two separate joints. They are labeled as *E* and *F*.

   A sliding joint connects the press head with the base. This joint is labeled as *G*.

4. ***Identify Any Points of Interest***

   Motion of the end of the handle is desired and is labeled as point of interest *X*.

5. ***Draw the Kinematic Diagram***

   The kinematic diagram is given in Figure 1.30.

6. ***Calculate Mobility***

   To calculate the mobility, it was determined that there are six links in this mechanism, as well as six pin joints and one slider joint. Therefore,

$$n = 6, j_p = (6 \text{ pins} + 1 \text{ slider}) = 7, j_h = 0$$

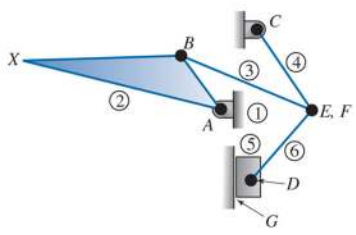


FIGURE 1.30 Kinematic diagram for Example Problem 1.8.

and

$$M = 3(n - 1) - 2j_p - j_h = 3(6 - 1) - 2(7) - 0 = 15 - 14 = 1$$

With one degree of freedom, the mechanical press mechanism is constrained. Moving only one link, the handle, precisely positions all other links in the press, sliding the press head onto the work piece.

EXAMPLE PROBLEM 3.3

Figure 3.6 shows a front loader. Use trigonometry to determine the required length of the cylinder to orient arm *AB* in the configuration shown.

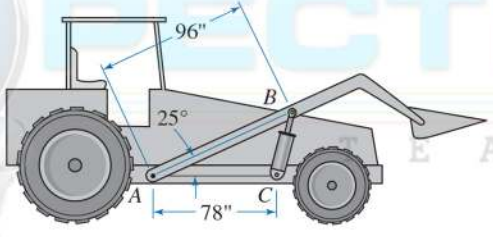


FIGURE 3.6 Front loader for Example Problem 3.3.

SOLUTION:

1. ***Determine Length* BC**

By focusing on the triangle created by points *A*, *B*, and *C*, it is apparent that this is a Case 3 problem. The third side can be found by using the law of cosines:

$$BC = \sqrt{AC^2 + AB^2 - 2(AC)(AB)\cos\angle BAC}$$

$$= \sqrt{(78\,\text{in.})^2 + (96\,\text{in.})^2 - 2(78\,\text{in.})(96\,\text{in.})\cos 25^\circ}$$

$$= 41.55\,\text{in.}$$

Because determining the remaining angles was not required, the procedure described for Case 3 problems will not be completed.

EXAMPLE PROBLEM 3.9

Three forces act on a hook as shown in Figure 3.17. Using the analytical component method, determine the net effect of these forces.

SOLUTION:

1. ***Use x-axis, Angle Method to Determine Resultant Components***

The horizontal and vertical components of each force are determined by trigonometry and shown in Figure 3.18. Also shown are the vectors rearranged in a tip-to-tail fashion. The components are organized in Table 3.1.

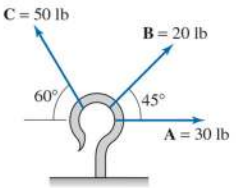


FIGURE 3.17 Forces for Example Problem 3.9.

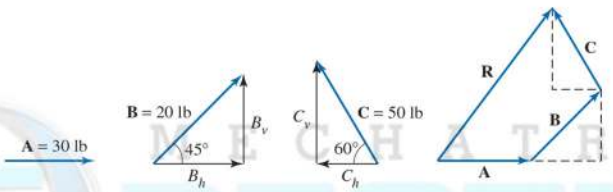


FIGURE 3.18 Components of vectors in Example Problem 3.9.

TABLE 3.1 Vector Components for Example Problem 3.9.

Vector	Reference Angle θ_x	*h* component (lb) $F_h = F\cos\theta_x$	*v* component (lb) $F_v = F\sin\theta_x$
A	0°	$A_h = (30)\cos 0° = +30$ lb	$A_v = (30)\sin 0° = 0$
B	45°	$B_h = (20)\cos 45° = +14.14$ lb	$B_v = (20)\sin 45° = +14.14$ lb
C	120°	$C_h = (50)\cos 120° = -25$ lb	$C_v = (50)\sin 120° = +43.30$ lb
		$R_h = 19.14$	$R_v = 57.44$

Notice in Figure 3.18 that adding the magnitudes of the horizontal components is tracking the total "distance" navigated by the vectors in the horizontal direction. The same holds true for adding the magnitudes of the vertical components. This is the logic behind the component method of combining vectors. For this problem, adding the individual horizontal and vertical components gives the components of the resultant as follows:

$$R_h = 19.14 \text{ lb.}$$

and

$$R_v = 57.44 \text{ lb.}$$

2. ***Combine the Resultant Components***

The resultant is the vector sum of two perpendicular vectors, as shown in Figure 3.19.

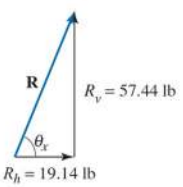


FIGURE 3.19 Resultant vector for Example Problem 3.9.

The magnitude of the resultant can be found from equation (2.13):

$$R = \sqrt{R_h^2 + R_v^2}$$
$$= \sqrt{(19.14\,\text{lb})^2 + (57.44\,\text{lb})^2} = 60.54\,\text{lb}$$

The angle of the resultant can be found:

$$\theta_x = \tan^{-1}\left(\frac{R_v}{R_h}\right) = \tan^{-1}\left(\frac{57.44\text{ lb}}{19.14\text{ lb}}\right) = 71.6°$$

Thus the resultant of the three forces can be formally stated as

$$\mathbf{R} = 60.54 \text{ lb.} \angle 71.6°$$